Atherosclerosis Pathway & Network Analysis

Jane Chidimma E. | Personal bioinformatics project

A paired analysis of public microarray data to examine expression differences, biological pathways and network patterns in atherosclerotic plaque and macroscopically intact carotid tissue.

64	32	114
microarray samples	paired patients	unique mapped DEGs

Data and analytical approach

GSE43292 contains RMA-preprocessed data from 32 hypertensive patients on platform GPL6244. The workflow checks metadata, fits a paired limma model, annotates genes, applies GO/KEGG enrichment, and examines a STRING functional-association network using Cytoscape.

Differential expression

The analysis identified 147 significant probe sets at BH-adjusted p < 0.01 and |log2FC| > 1. Removing missing and duplicate gene symbols leaves 114 genes: 58 higher and 56 lower in plaque.

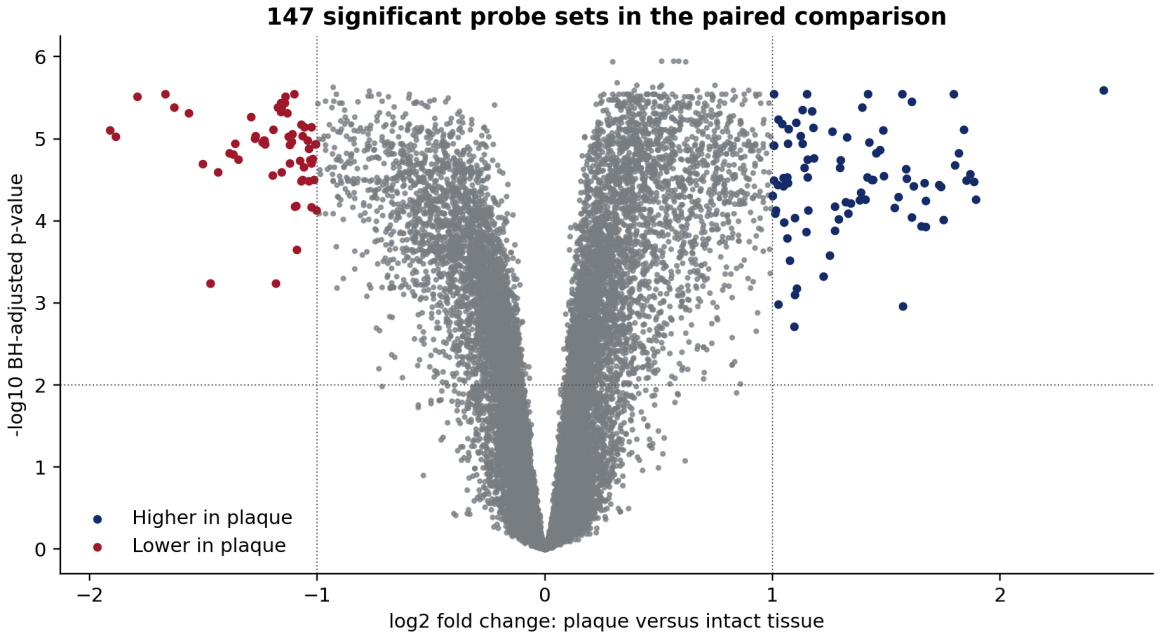


Figure 1. Differential expression across 33,297 probe sets. Coloured points meet both thresholds. Positive log2 fold change indicates higher expression in plaque.

Pathway enrichment

Two complementary approaches connect individual gene changes to biological processes. GSEA evaluates the full ranked gene list. ORA tests the selected DEG list against genes eligible for measurement and annotation.

Gene set enrichment analysis

GSEA using fgsea highlights immune-related processes toward the plaque end of the ranked list. At BH FDR < 0.05, 1,379 of 6,310 GO biological processes and 121 of 353 KEGG pathways are significant across both enrichment directions.

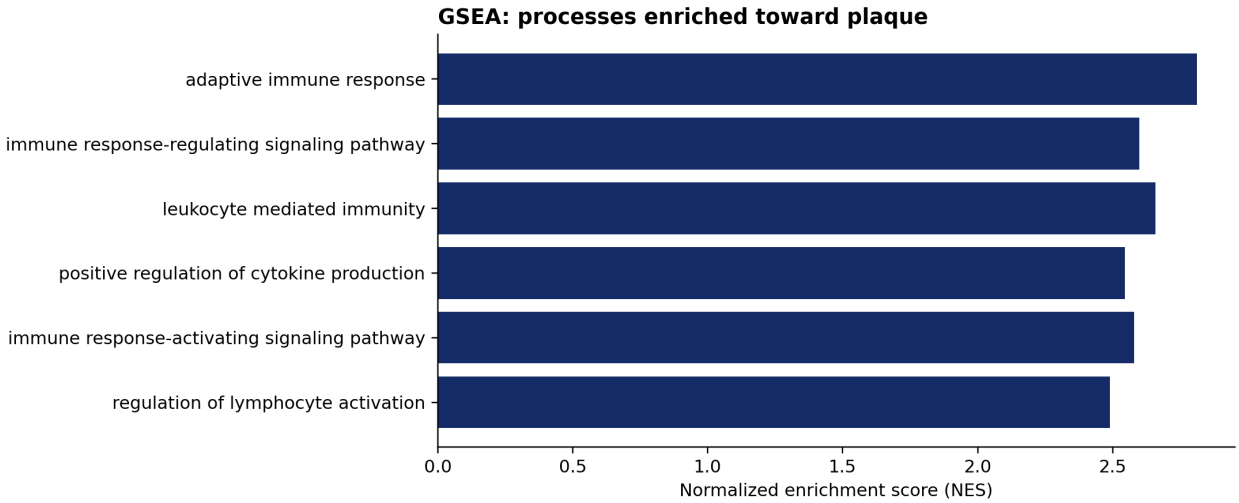


Figure 2. Six positive-NES GO biological processes selected by smallest adjusted p-value and displayed by NES. Input: 19,885 unique mapped Entrez IDs ranked by log2 fold change.

Over-representation analysis

ORA with a measured-gene background identifies 15 GO biological processes, including complement and immune processes. Two KEGG pathways pass BH FDR < 0.05:

KEGG pathway	DEG overlap	Adjusted p-value
Tryptophan metabolism	4	0.0303
PPAR signaling pathway	5	0.0303

Biological interpretation

The immune-related patterns support a shared interpretation across the methods. However, GSEA and ORA answer different questions. ORA alone does not establish whether a pathway is activated or suppressed.

Methods: gene sets contain 10-500 eligible genes. GSEA uses log2 fold-change ranking. ORA uses database-specific backgrounds of measured, mapped genes. Multiple-testing correction uses the Benjamini-Hochberg method. GO annotations: org.Hs.eg.db and GO.db 3.18.0. KEGG retrieval: 21 October 2026.

Network analysis & conclusions

The STRING functional-association network contains 47 nodes and 107 edges. Degree and maximal-clique centrality (MCC) identify highly connected genes using different ranking criteria. Nine genes appear in both top-ten rankings.

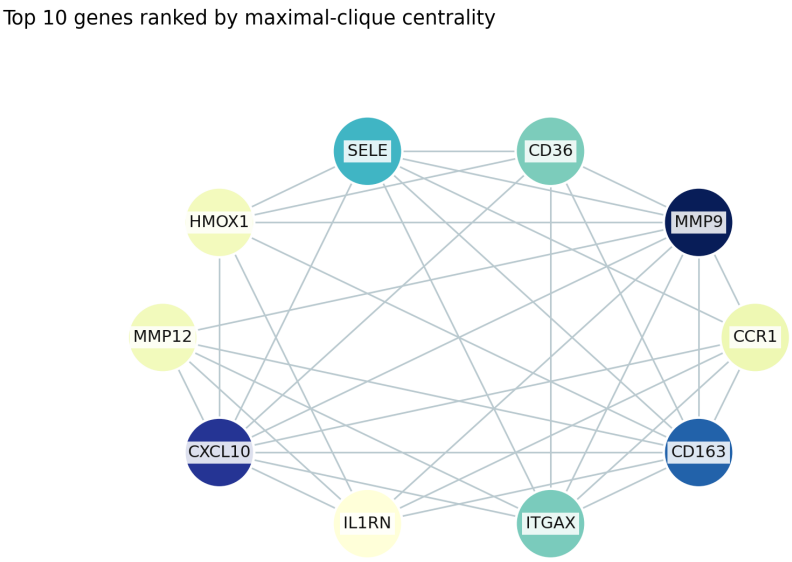


Figure 3. Induced subgraph of the ten highest MCC-ranked nodes. Edges represent STRING associations. Darker nodes have higher MCC scores. Connectivity does not provide independent biological validation.

Interpretation and limitations

The results support immune involvement in the plaque-versus-intact tissue comparison. A single observational dataset cannot establish causality or validated biomarkers. Cell composition may contribute to the patterns. Intact tissue from these patients does not represent healthy-donor tissue. The network analysis focuses on upregulated genes and reflects the STRING associations available for this project.

Project background and outputs

This personal project was developed from my MSc coursework in Network and Pathway Analysis using the public GSE43292 dataset. The project outputs include differential expression results, pathway enrichment summaries, network visualisations and documented methods.

Selected sources

NCBI Gene Expression Omnibus. Dataset GSE43292; platform GPL6244.

Yan et al. (2022). DOI: 10.1177/03000605211039480.

Ritchie et al. (2015). limma. DOI: 10.1093/nar/gkv007.

Subramanian et al. (2005). GSEA. DOI: 10.1073/pnas.0506580102.

Korotkevich et al. fgsea. DOI: 10.1101/060012.